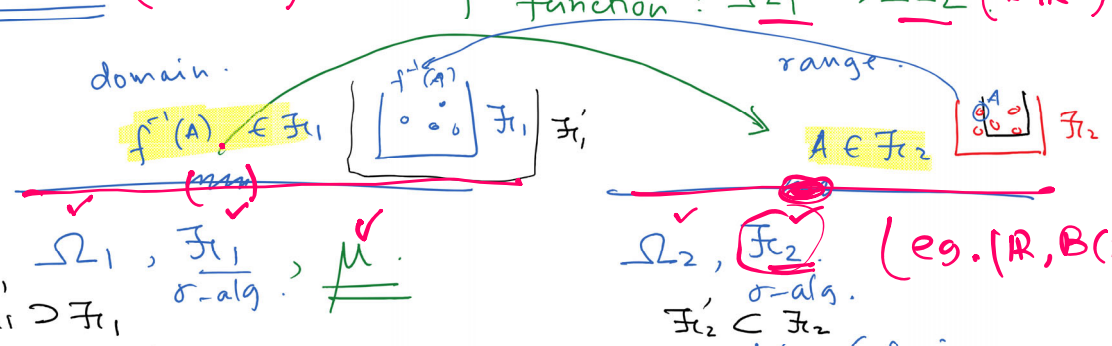
Notes C

Monday, September 28, 2026 9:24 AM

Sunday, September 20, 2026
8:43 PM

Assume $\Omega \neq \phi$, $\mathcal{F}$ σ-alg. $(\Omega, \mathcal{F})$ mble space.

Ch. 2. ("r.v's")



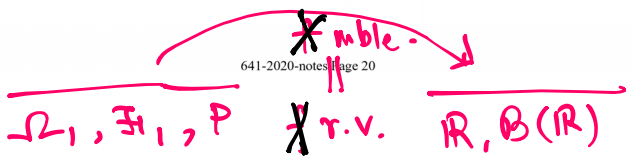
This function f is called measurable (f is $\langle \mathcal{F}_1, \mathcal{F}_2 \rangle$ - mble) if:

$$f^{-1}(A) \in \mathcal{F}_1 \quad \forall A \in \mathcal{F}_2$$

$f^{-1}(A) = \{\omega_1 \in \Omega_1 : f(\omega_1) \in A\}$

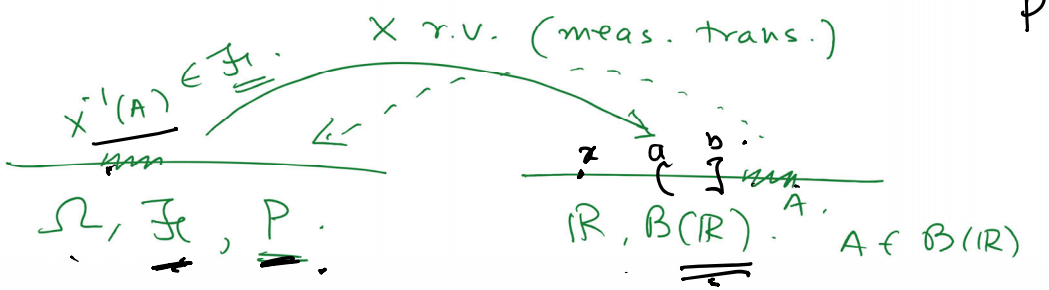
Note: If f is $\langle \mathcal{F}_1, \mathcal{F}_2 \rangle$ - mble then it is $\langle \mathcal{F}_1', \mathcal{F}_2' \rangle$ - mble for all $\mathcal{F}_1' \supset \mathcal{F}_1$ and $\mathcal{F}_2' \subset \mathcal{F}_2$.

If left side (domain) has a prob. meas, call it P, and right-side (range) has $(\Omega_2, \mathcal{F}_2) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ Then a measurable transformation is called a



$X \to X$

random variable, and is denoted as X, Y etc.
(instead of f, g etc.)



$P(X^{-1}(A)) = ?$

$A \in \mathcal{B}(\mathbb{R})$

Then $P(X \in A) = P(\{\omega : X(\omega) \in A\})$

$P(X > 0)$ "=" $P(X \in (0,\infty))$ $\equiv P(X^{-1}(A)) = P(X^{-1}(0,\infty))$

$X^{-1}(A) \in \mathcal{F}$.

$\longrightarrow$ can compute this.

$P(X \in (a,b]) = P(a < X \le b)$

X: r.v.

$= P(X^{-1}((a,b]))$

$\{\omega \in \Omega : X(\omega) \in (a,b]\}$

$\{X \in (a,b]\}$.

ex: (1) $f : (\Omega_1, \mathcal{F}_1) \longrightarrow (\Omega_2, \mathcal{F}_2)$, $c \in \Omega_2$

($\Omega_2 = \mathbb{R}$, $\mathcal{F}_2 = \mathcal{B}(\mathbb{R})$)

$f(\omega) = c$, $\forall\, \omega \in \Omega_1$

3.2.



take any $A \in \mathcal{F}_2$.

$$f^{-1}(A) = \{\omega \in \Omega_1 : f(\omega) \in A\},$$

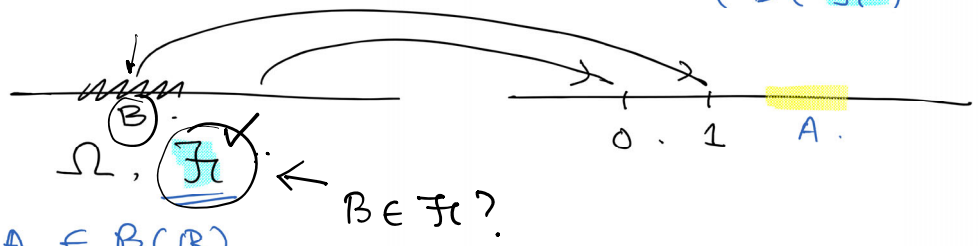
$$= \begin{cases} \phi & \text{if } c \notin A. \\ \Omega_1 & \text{if } c \in A. \end{cases}$$

$\in \mathcal{F}_1$ (since $\phi, \Omega_1 \in \mathcal{F}_1$).

So constant functions are always meas.

ex: (2) $f: (\Omega, \mathcal{F}) \longrightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$

$f(\omega) = 1_B(\omega)$, $B \subset \Omega$. ($B \in \mathcal{F}$)



$\forall A \in \mathcal{B}(\mathbb{R})$

$f^{-1}(A) = \{\omega \in \Omega : f(\omega) \in A\}$.

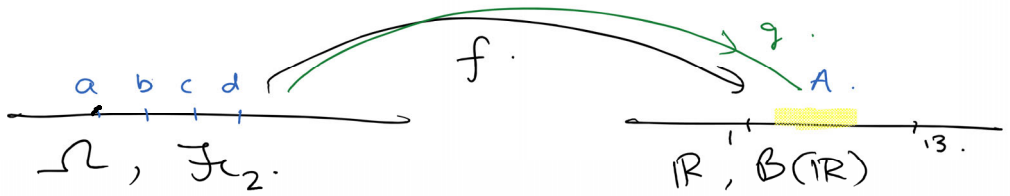
$$= \begin{cases} \phi & 0,1 \notin A \\ B^c & 0 \in A, 1 \notin A \\ B & 0 \notin A, 1 \in A \\ \Omega & 0,1 \in A \end{cases}$$

f is measurable iff $B \in \mathcal{F}$. ($\Rightarrow B^c \in \mathcal{F}$)

ex (3) mble transformations give "partition" of $\Omega_1 \to \mathcal{F}_1$ has be be a finer partion than that.

from ch.1.

$\Omega = \{a,b,c,d\}$. $\mathcal{F} \equiv \mathcal{F}_2 = \{\Omega, \phi, \{a\}, \{b,c,d\}\}$ $\{a\}^c$



If $f(a) = 1$, $f(b) = 13$, $f(c) = 13$, $f(d) = 13$.

Check that f is $(\Omega, \mathcal{F}_2)$ mble.

(here $f^{-1}(A) = \phi$ or Ω, $\{a\}$ or $\{b,c,d\}$)

If $g(a) = 1$, $g(b) = 2$, $g(c) = 13$, $g(d) = -10$

Then g is not $< \mathcal{F}_2, B(\mathbb{R}) >$ mble.

$g^{-1}(\{13\}) = \{c\} \notin \mathcal{F}_2$.

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ex (4) $\Omega = \{H, T\}$, $\mathcal{F} = \mathcal{P}(\Omega) = \{\phi, \Omega, \{H\}, \{T\}\}$

$X: \Omega \to \mathbb{R}$.

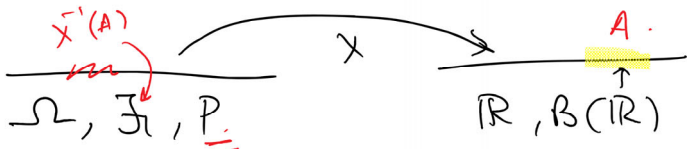
$H \mapsto 1$

$T \mapsto 0$.

P meas on $(\Omega, \mathcal{F})$

$P(\{H\}) = p$, $P(\{T\}) = q$.

$0 \le p, q \le 1$, $p+q=1$



$X = 1_{\{H\}}$, $\{H\} \in \mathcal{F}$ so X is a valid r.v.

$P(X=1) = P(\{\omega : X(\omega) = 1\}) = P(X^{-1}\{1\})$

$= P(\{H\}) = p$

Similarly $P(X=0) = q$.

This lets us talk about $P(X \in A) = P(X^{-1}(A))$

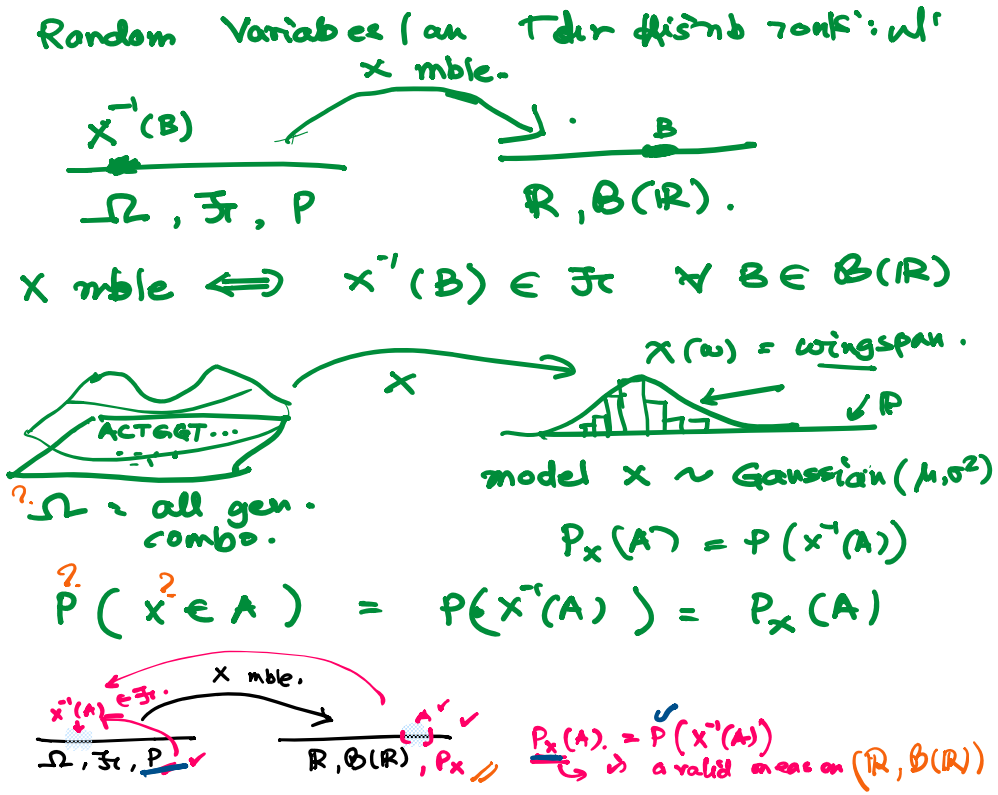
$P_X(A)$ ← a measure on the right side.

The Gaussian (or Normal) distribution, denoted as $X \sim \mathcal{N}(\mu, \sigma^2)$, appears ubiquitously in nature. This is not a mathematical coincidence, but the direct physical manifestation of the **Central Limit Theorem (CLT)**. Whenever a natural phenomenon is the cumulative sum of many independent, small-scale random variables, the resulting macroscopic variable naturally converges to a Gaussian distribution.

Here are the most robust examples in nature:

1. Quantitative Biological Traits (Biometrics)

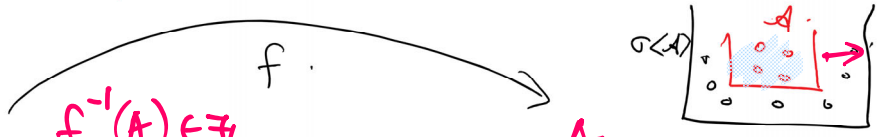
Traits like human height, blood pressure, and the wingspan of birds are classic examples. These are **polygenic traits**—they are not controlled by a single gene, but by the additive effects of hundreds of genetic variations, combined with numerous independent environmental factors (like nutrition). Because these countless micro-factors are added together, the resulting phenotype follows a normal curve.



$(P_X \text{ on } \mathbb{R}, \mathcal{B}(\mathbb{R}))$ → we usually don't know P, we specify P_X.

This measure is called induced measure (in stat, "distribution of X") → next section.

prop 2.1.1 (i)



f is measurable ($\langle \mathcal{F}_1, \mathcal{F}_2 \rangle$ mble) if

$f^{-1}(A) \in \mathcal{F}_1 \quad \forall \; A \in \mathcal{A}$. ($\mathcal{A} \subseteq \sigma\langle \mathcal{A} \rangle = \mathcal{F}_2$)

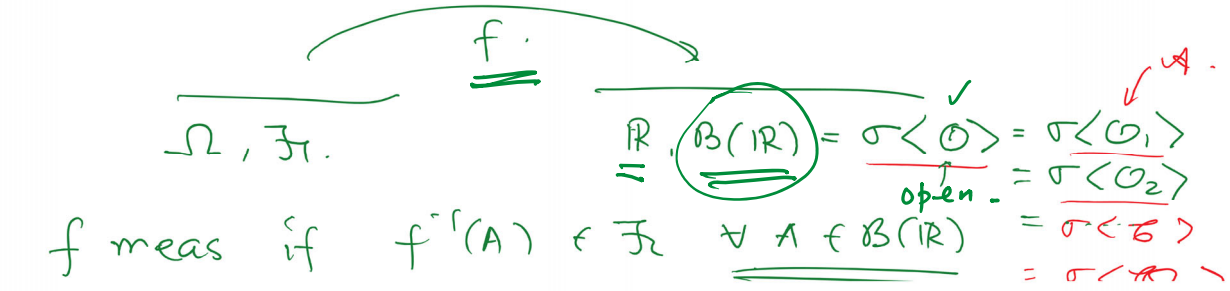
$\left(\overset{\text{prop 2.1.1 (i)}}{\Longrightarrow}\right) \quad f^{-1}(A) \in \mathcal{F}_1 \quad \forall A \in \mathcal{F}_2$

$T^{-1}(A) \cup T^{-1}(B)$

$= T^{-1}(A \cup B)$.

$\bigcup_{i=1}^{\infty} T^{-1}(A_i)$

$= T^{-1}\left(\bigcup_{i=1}^{\infty} A_i\right)$



f meas if $f^{-1}(A) \in \mathcal{F} \quad \forall \; A \in \mathcal{B}(\mathbb{R})$

$\mathcal{O}_1 = \{(a,b), a<b, a,b \in \mathbb{R}\}$

$\mathcal{O}_2 = \{[a,b)\}$.

⋮

$$\mathcal{F} = \{ A \in \mathcal{F}_2 : T^{-1}(A) \in \mathcal{F}_1 \}$$

$\mathcal{A} \subseteq \mathcal{F} \subseteq \mathcal{F}_2$. and $\mathcal{F}$ σ-alg. (pf)

Since $\mathcal{F}$ is σ-alg cont $\mathcal{A}$, and $\sigma\langle\mathcal{A}\rangle$ is smallest such, $\sigma\langle\mathcal{A}\rangle = \mathcal{F}_2 \subseteq \mathcal{F} \subseteq \mathcal{F}_2$.

From prop ~~2.3.1~~ 2.1.1 (i): $f : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is measurable if -

$\sigma\langle\mathcal{O}\rangle = \sigma\langle\mathcal{O}_1\rangle = \mathcal{B}(\mathbb{R})$

$\Omega, \mathcal{F}$ — $\mathbb{R}, \mathcal{B}(\mathbb{R})$ — $\mathcal{O}$

$f^{-1}(O) \in \mathcal{F} \quad \forall \; O \in \{\text{open sets of } \mathbb{R}\}$ -xx

or $f^{-1}((a,b)) \in \mathcal{F} \quad \forall \; (a,b] \in \mathcal{O}_1$ xx

in particular, f is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$-measurable if

$$f^{-1}((-\infty, a]) \in \mathcal{F} \quad \forall\, a \in \mathbb{R}.$$

$$\| \quad \{\omega \in \Omega : f(\omega) \in (-\infty, a]\} = \{\omega \in \Omega : f(\omega) \le a\}$$

or $f^{-1}([b, \infty)) \in \mathcal{F} \quad \forall\, b \in \mathbb{R}$ ✓

$$\| \quad \{f \ge b\} = \{\omega \in \Omega : f(\omega) \ge b\}$$

So X is r.v if

$$\{X \le a\} = \{\omega : X(\omega) \le a\} \in \mathcal{F} \quad \forall\, a \in \mathbb{R}.$$

Prop 2.1.2:

$(\mathbb{R}^p \to \mathbb{R}^k)$ $p=k=1$ $\Rightarrow$

$f : (\mathbb{R}, \mathcal{B}(\mathbb{R})) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$

Ω_1 $\mathcal{F}_1$ $\sigma\langle\mathcal{O}\rangle$ Ω_2 $\mathcal{F}_2$

$\mathcal{O}$ = {all open sets}

f is mble if $f^{-1}(O) \in \mathcal{B}(\mathbb{R})$ $\forall O \in \mathcal{O}$. ($\mathcal{B}(\mathbb{R}) = \mathcal{F}_1$)

If f cont $f^{-1}(O)$ is an open set. (HW 1)

f meas if $f^{-1}(A) \in \mathcal{F}$ $\forall A \in \mathcal{B}(\mathbb{R})$ $= \sigma\langle \mathcal{C} \rangle$
$= \sigma\langle \mathcal{O}_3 \rangle$
$= \sigma\langle \mathcal{O}_4 \rangle$
$= \vdots$

$\mathcal{B}(\mathbb{R}) = \sigma\langle \{(-\infty, a] \cdots\} \rangle$.

f is meas (by prop. 2.1.1 (i)) if

$f^{-1}((-\infty, a]) \in \mathcal{F}$.

$\{\omega \in \Omega : f(\omega) \in (-\infty, a]\}$

$\{\omega \in \Omega : f(\omega) \leq a\}$.

$\mathcal{B}(\mathbb{R})$ is also $\sigma\langle \{\text{open sets}\} \rangle$

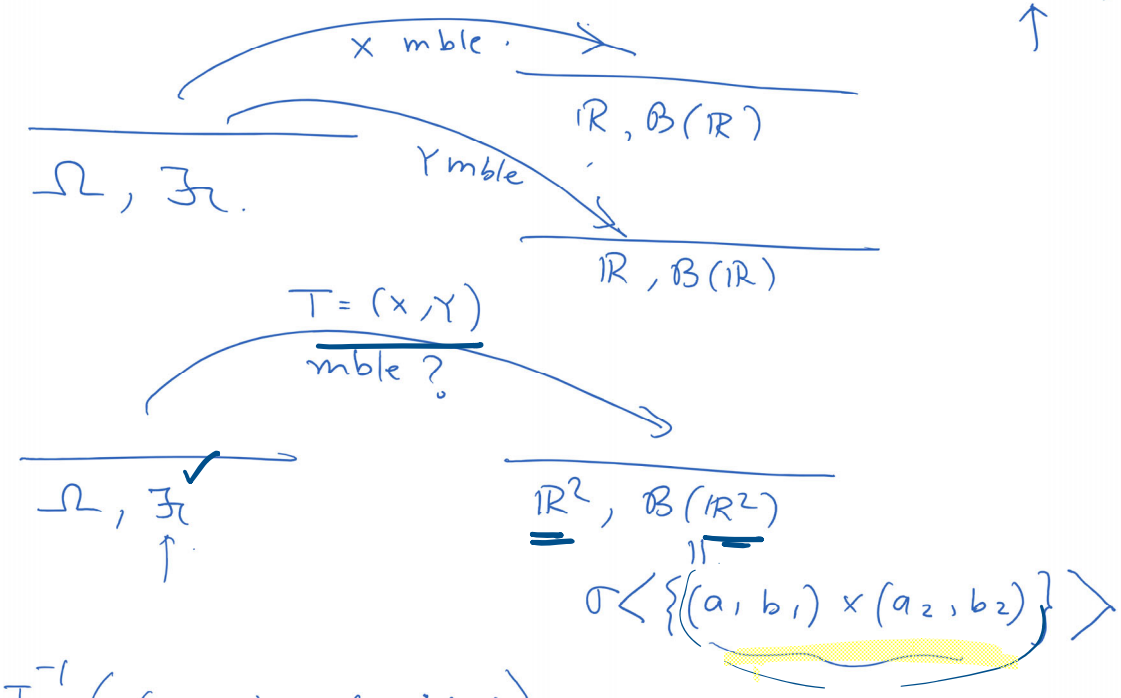
if $f^{-1}(\mathcal{O}) \in \mathcal{B}(\mathbb{R})$, for any open set $\mathcal{O}$
Then f is $\langle \mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}) \rangle$ mble.

In particular, if f is cont,
Then $f^{-1}(\mathcal{O})$ is cont.

i.e $f^{-1}(O) \in \mathcal{O} \subseteq \sigma\langle\mathcal{O}\rangle \equiv \mathcal{B}(\mathbb{R})$

This shows all cont functions from $\mathbb{R} \to \mathbb{R}$ are $\langle \mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R})\rangle$ mble.

prop. 2.1.3 (i) : (using X, Y notation instead of f_1, f_2 $(K=2)$)



$T^{-1}((a_1, b_1) \times (a_2, b_2)) \in \mathcal{F}$?

$$T^{-1}\left((a_1,b_1)\times(a_2,b_2)\right) =$$

$$= \{\omega : T(\omega) \equiv (X(\omega), Y(\omega)) \in (a_1,b_1)\times(a_2,b_2)\}$$

$$= \{\omega : a_1 < X(\omega) < b_1\} \cap \{\omega : a_2 < Y(\omega) < b_2\}$$

$\in \mathcal{F}$ ✓ (since X mble) $\qquad \in \mathcal{F}$.

$\in \mathcal{F}$.

$$Z = X + Y = g(T),$$

$T : \Omega, \mathcal{F} \to \mathbb{R}^2, \mathcal{B}(\mathbb{R}^2)$

$g(t) = t_1 + t_2$ when $t = (t_1, t_2) \in \mathbb{R}^2$.

$g : \mathbb{R}^2 \to \mathbb{R}$.

Here g is cont. map. (hence mble)

and $T = (X, Y)$ is mble.

and comp. of 2 mble maps is mble.

$\Rightarrow Z = X + Y : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is mble.

Let X, Y be measurable, then $Z = \max(X,Y)$ $W = \min(X,Y)$ are meas.

$Z(\omega) = \max(X(\omega), Y(\omega))$ $\forall\, \omega \in \Omega$

$\{Z \le a\} = \{\omega : \max(X(\omega), Y(\omega)) \le a\}$

$= \{\omega : X(\omega) \le a\} \cap \{Y \le a\} \in \mathcal{F}$.

$\in \mathcal{F}$ $\in \mathcal{F}$.

$\min(X,Y) = -\max(-X, -Y)$ is meas.

$A = \{X = Y\} = \{\omega : X(\omega) = Y(\omega)\}$.

$= \{(X-Y) = 0\} = (X-Y)^{-1}(\{0\}) \in \mathcal{F}$.

meas.

$= \{\omega : (X(\omega) - Y(\omega)) = 0\}$

$B = \{X > Y\}$ ← mble.

mble $\in \mathcal{B}(\mathbb{R})$

$= \{\omega : (X(\omega) - Y(\omega)) > 0\} = (X-Y)^{-1}((0,\infty)) \in \mathcal{F}$

Limits:

(extended reals).

$+\infty \geq x \quad \forall x \in \mathbb{R}$.

$\mathcal{B}(\bar{\mathbb{R}}) = \sigma\langle \mathcal{B}(\mathbb{R}) \cup \{+\infty\} \cup \{-\infty\} \rangle$

(a) (b)

problem 2.4 (check)

one elt.

$(-a,\infty) \cup \{+\infty\} = (a,\infty]$

$f: (\Omega, \mathcal{F}) \longrightarrow (\bar{\mathbb{R}}, \mathcal{B}(\bar{\mathbb{R}}))$

← extend real valued f

measurable if

$\forall a \in \mathbb{R}, \{\omega : f(\omega) \leq a\} \in \mathcal{F}$.

← same prop- for mbl-ility of real valued f.

$\left[\{f = \infty\} = \bigcap_{n \geq 1} \{f > n\} \in \mathcal{F}. \text{ check} \right]$

$\omega \in f^{-1}(\{\infty\})$

$f^{-1}(\{-\infty\}) = \{\omega : f(\omega) = -\infty\}$

$= \{\omega : f(\omega) \leq -n, \forall n \geq 1\}$

$= \bigcap_{n \geq 1} f^{-1}((-\infty, -n]) \in \mathcal{F}$.

Let $X : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be measurable.

($\mathcal{G} \subseteq \mathcal{F}$. $\mathcal{B}$.)

$\mathcal{H} \subsetneq \mathcal{G} = \{ X^{-1}(B) : B \in \mathcal{B}(\mathbb{R}) \} \subseteq \mathcal{F}$

easy to check it is a σ-alg.

this is called $\sigma\langle X \rangle$ → smallest σ-alg that makes X mble
or, σ-alg generated by X.

Take X, Y mble.

$$\sigma\left\langle \{ X^{-1}(B) : B \in \mathcal{B}(\mathbb{R}) \} \cup \{ Y^{-1}(C) : C \in \mathcal{B}(\mathbb{R}) \} \right\rangle$$

not a σ-alg, in general.

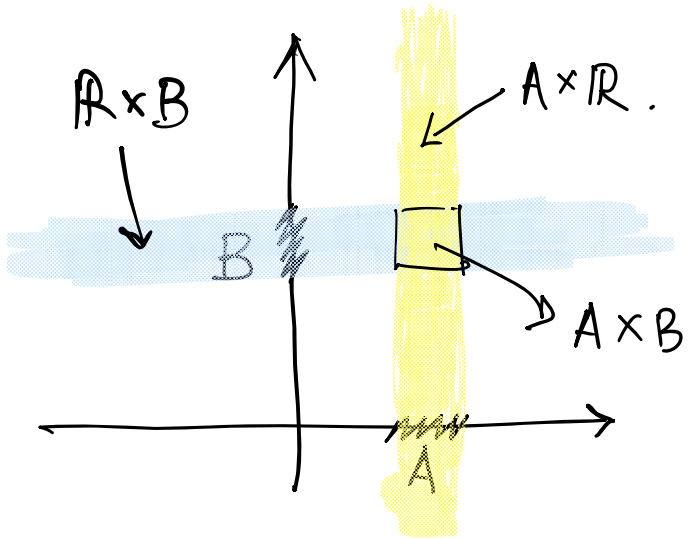
$= \sigma\langle X, Y \rangle$ → σ-alg gen. by X, Y
(smallest σ-alg that makes both X, Y mble)

Example 2.1.4

$f_i : \mathbb{R}^2 \to \mathbb{R} \quad : \quad f_i(x_1, x_2) = x_i$.

then $f_1^{-1}(A) = A \times \mathbb{R}$.

$f_2^{-1}(B) = \mathbb{R} \times B$.



$\Rightarrow \quad A \times B = f_1^{-1}(A) \cap f_2^{-1}(B)$

$\mathcal{B}(\mathbb{R}^2) = \sigma\langle \quad \rangle$

$\sigma\langle \{ f_1^{-1}(A), f_2^{-1}(B), A, B \in \mathcal{B}(\mathbb{R}) \} \rangle$

$\sigma\langle \{ f_\alpha : \alpha \in [0,1] \} \rangle = \sigma \{ \{ f_i : i \in \text{cftble set} \} \rangle$